

Library – Book Issue & Return (Feature Set 4)

Assignment Prompt

Design and implement a simple library management system for the process of issuing and returning books. The system should include the following features (Feature Set 4):

1. **Student Details** – Maintain student information such as Student ID, Name, Phone, Email and Course.
2. **Book Details** – Maintain book information such as Book ID, Title, Author, Category and Availability Status.
3. **Book Search** – Allow users to search for books by title, author or category.
4. **Book Availability** – Check whether the book is available or not.
5. **Book Renewal** – Allow students to renew issued books (if eligible).

Prepare the following three components for the above system:

1. ER Diagram

Design an Entity–Relationship (ER) diagram for the library system. Identify the main entities, attributes, relationships and cardinalities based on the features given above. Include the following entities and attributes at least:

- **Student** (Student_ID, Name, Phone, Email, Course)
- **Book** (Book_ID, Title, Author, Category, Availability_Status)
- **Issue_Return** (Issue_ID, Issue_Date, Due_Date, Return_Date, Status)
- **Renewal** (Renewal_ID, Renewal_Date, New_Due_Date)

Show the relationships between the entities with proper cardinalities (e.g., one to many, one to one, etc.).

2. Flowchart

Prepare a flowchart to show the process of book issue, return and renewal. The flowchart should cover the following:

- Enter and validate student details.
- Search for a book (by title, author or category).
- Check book availability.
- Issue the book and update the records.
- Handle book return and calculate fine (if any).
- Allow book renewal (if eligible) and update the due date.

3. Algorithm

Write the algorithm for the book issue, return and renewal process. The algorithm should include steps for:

- Validating student details
- Searching and checking book availability
- Issuing a book
- Returning a book (with fine calculation if late)
- Renewing a book (with eligibility check and due date update)

Use clear steps and proper logic (if–else conditions, loops if required).

Assignment Requirements: Library System

1. Objective:

Design and document the logical flow and data structure for a Library Book Search, Availability, and Renewal system.

2. Deliverable 1: Algorithm:

Create a precise, step-by-step numbered algorithm detailing the complete process from Student ID entry to the final book renewal or system termination, including all validation checks and conditional decisions as defined in the specification.

3. Deliverable 2: Flowchart:

Construct a detailed, visual flowchart that accurately diagrams the full process flow of the algorithm. All start/stop points, input steps, processing steps, and decision points must be represented using standard flowchart symbols and logical connectors.

4. Deliverable 3: ER Diagram:

Design a comprehensive Entity-Relationship (ER) Diagram that models the database for this system. This diagram must include all necessary entities (e.g., STUDENT, BOOK), their attributes (with primary keys identified), and the precise relationships (cardinality and participation) between them, as required to support the defined algorithm.

5. Submission Format:

All three deliverables (Algorithm, Flowchart, and ER Diagram) must be clearly labeled, professionally presented, and integrated into a single cohesive assignment document.